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TOILET

Abstract

The present invention relates to a toilet (1, 2, 3, 4, 5) with a toilet bowl (1) having a toilet interior (12), having a mounting (4) which is preferably arranged in a fixed manner on the toilet bowl (1), having a toilet seat (2) which is arranged on the mounting (4) such that it can be moved, preferably pivoted, particularly preferably vertically pivoted, between a use position and a non-use position, or is formed by the toilet bowl (1), and having a toilet lid (3) which is formed and arranged on the mounting (4), such that it can be moved, preferably pivoted, particularly preferably vertically pivoted, in order to close or release the toilet interior (12) by moving, preferably pivoting. The toilet (1, 2, 3, 4, 5) is characterised by at least one fan (33, 44), which is designed and arranged to generate a circulating air flow in the toilet interior (12) of the toilet bowl (1), which is preferably closed by means of the toilet lid (3), and by at least one ioniser (34, 46), preferably at least one ozone generator (34, 46), which is designed and arranged to generate ionised air, preferably ozone, and to dispense it into the circulating air flow.

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Background/Summary

[0001] The present invention relates to a toilet.

[0002] Nowadays it is common to use toilets, which can be used by one person sitting as the user. A toilet is a sanitary device for holding bodily excretions such as, in particular, feces and urine. A toilet usually has a toilet bowl or basin, which forms an interior space into which bodily excretions can be placed. For this purpose, the person can sit on a toilet seat of the toilet, which is designed as a separate component of the toilet that can be pivoted relative to the toilet bowl and has its own seat surface. Alternatively, the upper rim of the toilet bowl can also serve as the seat of the toilet.

Typically, bodily excretions can be discharged by a water flush through a drain to a sewer system.

[0003] In addition to bodily excretions, gases can also be emitted by the person using the toilet or be produced by the bodily excretions, which can enter the ambient air in the immediate vicinity of the toilet and cause an odor nuisance. Moreover, the toilet interiors, i.e., the inside of the toilet bowl, are to be cleaned regularly.

[0004] For automatic cleaning of interiors of toilets or toilet bowls, the use of so-called toilet flushers is known, which are hung on the toilet rim in the toilet interior, such that when the water flush is operated, the toilet flushers are surrounded by water and a part of a cleaning agent is emitted into the water. This allows the toilet interior to be cleaned to a certain extent where the water mixed with cleaning agent can reach it. The cleaning agent can also contain a fragrance, which can be released and emitted into the ambient air. Both of these can take place automatically whenever the water flush is operated.

[0005] The disadvantage of this is that the cleaning effect can only occur in the toilet interior where water mixed with the cleaning agent reaches it. This can result in uneven cleaning of the surface of the toilet interior. Moreover, only the toilet interior itself can be cleaned in this manner. In particular, the toilet seat and especially its seat surface are not reached at all by this type of cleaning.

[0006] A further disadvantage is that any odor nuisance that occurs can merely be masked by the scent of the cleaning agent, but itself persists.

[0007] The disadvantage in each case is that the effect of the cleaning and the fragrance depends on the quantity of cleaning agent that is dissolved or released by the water of the water flush.

Accordingly, as the cleaning agent dwindles, the effects may also diminish.

[0008] The overall disadvantage is that the cleaning agent is consumed by the use of the toilet and must be replaced or refilled. This can lead to cost and effort for the user. In particular, the functions may not be available if or as long as the cleaning agent is used up.

[0009] An object of the present invention is to provide a toilet so that cleaning and in particular disinfection of a toilet, in particular the seat surface of the toilet seat of a toilet, can be improved and/or simplified. Additionally or alternatively, odor nuisance during and/or after using a toilet should be avoided or at least reduced. This should be as simple, automatic, robust, compact and/or cost-effective as possible. At least an alternative to the corresponding known options is to be created.

[0010] The object is achieved according to the invention by a toilet, by a toilet seat, by a toilet lid, by a mounting and by a fragrance unit having the features of the independent claims. Advantageous further developments are described in the subclaims.

[0011] Thus, the invention relates to a toilet with a toilet bowl with a toilet interior, with a mounting, with a toilet seat which is arranged on the mounting so as to be movable between a position of use and a position of non-use or is formed by the toilet bowl, and with a toilet lid which is designed and arranged to be movable on the mounting so as to close or release the toilet interior by moving it.

[0012] Such toilets are well known, commercially available and widely used. Depending on the design or application, the toilet bowl can be made of ceramic. In this case, the use of toilet seats made of plastic or wood is common. However, there are also toilet bowls made of metal which do not have a separate toilet seat but instead form the seat for the user themselves. If the toilet seat is designed separately, the toilet seat can be moved by the user away from the edge of the toilet bowl by hand, which represents the non-use position, since the toilet seat cannot be used by the user to sit on the toilet in this position. This non-use position can be used in particular for cleaning the toilet bowl including the toilet bowl rim. In the use position, the toilet bowl rim is covered by the toilet seat and the user can use the toilet seat in this position for sitting.

[0013] In any case, it is known and common practice to provide a movable toilet lid, which can be moved by the user to open and close the toilet interior by hand and can usually be pivoted or folded vertically. When the toilet lid is open, the toilet seat can be used by the user in the usage position. The toilet seat can also be moved with the toilet lid open as described above. To close the toilet lid, the toilet seat must be placed in the use position. When the toilet lid is closed, the toilet seat and the toilet interior are covered from above.

[0014] The toilet according to the invention is characterized by at least one fan, which is designed and arranged to generate a circulating air flow within the toilet interior of the toilet bowl, and by at least one ionizer, which is designed and arranged to generate ionized air and to deliver it to the circulating air flow.

[0015] The fan can be implemented, for example, as a ventilator or similar, and in any event serves to move air. To do this, the fan can suck in air from the toilet interior, pump it through itself and release this air back into the toilet interior as an air stream, so that a circulating air stream can be generated within the toilet interior or through the toilet interior. This allows the air present inside the toilet to be specifically conveyed and ionized, so that the surface, particularly the interior of the toilet, can be cleaned and/or disinfected using the ionized air of the circulating air flow. At the same time, odors can be removed by ionizing the air in the circulating air stream.

[0016] The interior of the toilet substantially represents the volume enclosed by the toilet bowl or its bowl body, which is open at the top. However, within the meaning of the present invention, additional areas can also be assigned to the toilet interior, which are jointly enclosed by the toilet bowl, the mounting, the toilet seat and/or the toilet lid when the toilet lid is closed. The toilet interior can therefore also include, in particular, the areas between the toilet bowl rim and the toilet seat or toilet lid and/or between the toilet seat and the toilet lid.

[0017] The air in the air stream is ionized by the ionizer using high voltage. Using an appropriate high voltage generator, the toilet itself is provided with the high voltage required to operate the ionizer. The ionized air or the generated radicals of the ionized air may cause chemical degradation processes, which can be used for disinfection and/or odor elimination.

[0018] Thereby, using an ozone generator as an ionizer can be one way to achieve the previously described ionization of the air.

[0019] In any event, an operation of at least the fan and the ionizer can take place by means of a control unit of the toilet, which can perform the corresponding functions and perform further functions such as, for example, the evaluation of sensor information, the activation of the functions according to time sequences or at predetermined points in time and similar, as will be described in more detail below. The required electrical power supply for the toilet and in particular its electrical and electronic components can be provided by an integrated electrical energy storage device, preferable a rechargeable one, as will be described in more detail below. Alternatively, the toilet

can be connected to an electrical supply network by cable or wirelessly, for example inductively. [0020] In any case, a toilet according to the invention can be designed to generate an ionized air flow and to circulate it within the toilet interior so that odors can be extracted from the toilet interior. The odors of the sucked-in air can then be eliminated or at least reduced by ionizing the sucked-in air and/or mixing it with ionized air. The excess ionized air can be pumped back into the toilet bowl with reduced or completely eliminated odors to reach the surfaces of the toilet interior and to clean or disinfect them. This can eliminate or at least reduce the odor nuisance for the surrounding area that can arise when using a toilet. At the same time, the interior of the toilet can be cleaned and/or disinfected. This makes it possible to achieve the simplest and/or most complete automatic cleaning and/or disinfection of the toilet interior.

[0021] For this purpose, at least one fan with an ionizer can be used, which can be arranged anywhere within the toilet in order to be able to carry out the functions described above. Thus, the fan and ionizer can be located in or on a component of the toilet such as the toilet bowl, toilet seat, toilet lid or toilet mounting. The fan and the ionizer can be located in or on the same component of the toilet or on different components. It is also possible to use several fans with one ionizer or one fan with several ionizers combined. However, several fans with several ionizers can also be used. If several fans and/or several ionizers are provided, thus individual fans and/or ionizers may be operated at the same time. If several fans are operated simultaneously, a circulating air flow can be generated in each case, whereby the circulating air flows can also complement, overlap or mix with one another. The extent to which several circulating air flows complement, overlap or mix can be influenced by the design and arrangement of the fans or their air inlets and/or their air outlets as well as by the strength or speed of the fans. All this can increase the design freedom and the possible uses in order to be able to provide a toilet according to the invention.

[0022] Preferably, at least one neutralizer may also be arranged in or on a component of the toilet in order to be able to neutralize excess ionized air, as will be described in more detail below. Multiple neutralizers can also be used.

[0023] Preferably, the mounting is fixedly mounted on the toilet bowl. This can enable a secure fit of the mounting, relative to which the toilet seat and/or the toilet lid can be moved.

[0024] The mobility of the toilet seat and/or the toilet lid can be adjusted as desired in order to implement the functions described above. Being able to pivot the toilet seat and/or the toilet lid relative to the toilet bowl can simplify implementation, as comparatively simple mechanical means such as hinges and similar can be used for this purpose. Pivoting can be done both horizontally and vertically, which can increase the design options. Implementing the pivoting preferably vertically, for example by means of a pair of hinges, can enable a particularly simple implementation. In this case, the swiveling can also take place directly above the toilet interior, so that the space required for this can be kept to a minimum.

[0025] Closing the toilet interior using the toilet lid when the air flow is circulating can particularly promote the cleaning or disinfecting effect of the circulating air flow of ionized air, since the ionized air is kept as far as possible within the toilet interior and can therefore exert its effect there. This also helps keep ionized air away from the area around the toilet, which could otherwise be perceived by the user as an odor nuisance.

[0026] Implementing the ionizer as an ozone generator can be advantageous in that the circulating air stream can be mixed with ozone in this way in order to achieve the effects described above using ozone.

[0027] In any case, the operation of the fan and the ionizer, preferably and/or the fan and a neutralizer, can preferably begin, be interrupted or end at least at a predetermined point in time, after expiration of at least a predetermined period of time and/or in dependence on a sensor-detected event, as will be described in more detail below. In this manner, in particular, the function of cleaning and/or disinfection by means of ionized air can be used in a time-controlled or time-delayed manner. On the one hand, cleaning and/or disinfection can be carried out automatically at

predetermined times in order to ensure regular cleaning and/or disinfection. Additionally or alternatively, cleaning and/or disinfection can always take place after a use of the toilet, which can be done automatically after the expiration of a predetermined time period after the end of the use of the toilet, preferably limited to a predetermined period of time.

[0028] The end of the use of the toilet can be detected by sensors comparable to the start of the use of the toilet as described before. When using a motion sensor, the end of the use of the toilet can be detected by the absence of the person. When using a pressure sensor, the relief of the pressure sensor can indicate the end of the use of the toilet. In any event, this can ensure that the toilet is cleaned and/or disinfected automatically after each use. This also applies to the odor reduction function that this also achieves.

[0029] Preferably, the operation of the fan and the ionizer, preferably and/or the fan and a neutralizer, can end automatically after a predetermined period of time. Such predetermined period of time can be suitably selected to ensure, on the one hand, the corresponding effect of the respective function and, on the other hand, to keep the operating time and the electrical energy consumption of the toilet device as low as possible.

[0030] According to one aspect of the invention, at least the toilet seat has a fan and an ionizer, preferably an ozone generator. This allows the previously described properties and functions to be implemented solely or partially by the toilet seat.

[0031] According to a further aspect of the invention, at least the toilet seat has a neutralizer, preferably a catalyst, particularly preferably an ozone catalyst, and the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet seat or the fan and the neutralizer of the toilet seat.

[0032] By means of the neutralizer, the excess ionized air or the excess ozone that was not used for odor neutralization and/or cleaning and/or disinfection can be removed from the circulating air flow in order to avoid odor nuisance to the user and/or the environment caused by the excess ionized air or the excess ozone. For this purpose, the circulating air flow can also be passed through the neutralizer, the catalyst or the ozone catalyst so that excess ionized air can be broken down by the neutralizer and/or the catalyst and/or excess ozone can be broken down by the ozone catalyst and thus kept away from the area around the toilet. This can prevent ionized air or ozone from entering the surrounding area of the toilet device, since the smell of the ionized air or ozone could be perceived as unpleasant by a user. Moreover, ionized air and/or ozone can be harmful to health and the surrounding area.

[0033] Accordingly, the fan and the ionizer of the toilet seat can be operated in combination with one another to generate a circulating air flow of ionized air as described above, which can clean and/or disinfect and/or reduce odors within the toilet interior. Alternatively, the fan and toilet seat neutralizer can be operated in combination to neutralize excess ionized air as previously described. In particular, the fan and ionizer can be operated first and then the fan and the neutralizer.

[0034] According to a further aspect of the invention, the toilet seat has at least one air inlet for sucking in the circulating air flow by means of the fan, and the toilet seat has at least one air outlet for discharging the circulating air flow by means of the fan. This allows air from the fan to be sucked in from the toilet interior via the air inlet into the toilet seat and be released back into the toilet interior via the air outlet to create the circulating air flow as previously described. For this purpose, several air inlets and/or several air outlets can be used, which can increase the design flexibility. In particular, multiple air inlets and/or multiple air outlets can enable or cause a more even distribution of air within the toilet interior, which can improve the uniform cleaning, disinfecting and/or odor-reducing effect of the ionized air.

[0035] Preferably aligning the air inlet horizontally to the toilet interior, i.e. in the horizontal plane towards the opening of the toilet seat, can be advantageous in order to allow the intake of air precisely from there and thus centrally.

[0036] Additionally or alternatively, aligning the air outlet preferably towards the toilet lid, i.e.

vertically upwards towards the inside of the toilet lid, can be advantageous in that the escaping ionized air from the toilet lid can be directed back or redirected towards the seat of the toilet seat, which can promote cleaning and/or disinfection of the seat surface of the toilet seat with ionized air from the circulating air flow.

[0037] Preferably, arranging the air inlet and the air outlet of the toilet seat as far apart from one another as possible can be advantageous in that this can increase the distance that the air outside the toilet seat has to travel between the air outlet and the air inlet. This allows the circulating air flow to flow through a larger part of the toilet interior, which can improve the cleaning, disinfection and/or odor-reducing effect of the ionized air.

[0038] According to a further aspect of the invention, at least the toilet lid has a fan and an ionizer, preferably an ozone generator. This allows the previously described properties and functions to be implemented solely or partially by the toilet lid.

[0039] According to a further aspect of the invention, at least the toilet lid has a neutralizer, preferably a catalyst, particularly preferably an ozone catalyst, and the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet lid or the fan and the neutralizer of the toilet lid. This allows the previously described properties and functions to be implemented solely or partially by the toilet lid.

[0040] According to a further aspect of the invention, the toilet lid has at least one air inlet for sucking in the circulating air flow by means of the fan, and the toilet lid has at least one air outlet for discharging the circulating air flow by means of the fan. This allows the previously described properties and functions to be implemented solely or partially by the toilet lid.

[0041] In this case too, the air inlet can preferably be aligned horizontally towards the toilet interior, which can bring about the advantages already described above. Preferably directing the air outlet towards the toilet seat, particularly preferably towards the seat surface, can have a corresponding cleaning and/or disinfecting effect on the toilet seat or its seat surface, as already mentioned above. This applies accordingly to the advantages of preferably arranging the air inlet and the air outlet of the toilet lid as far apart from one another as possible.

[0042] According to a further aspect of the invention, the toilet lid has a plurality of air outlets which are directed towards the toilet seat. This can improve the cleaning and/or disinfecting effect of the circulating air flow for the toilet seat or its seat surface, since the ionized air of the circulating air flow can be released from the toilet lid at several points directed at the toilet seat or its seat surface. This can be further improved by preferably arranging the plurality of air outlets in an evenly distributed manner.

[0043] According to a further aspect of the invention, the fan is arranged centrally in the toilet lid, directly connected to the air inlet and connected to at least the air outlet, preferably to the plurality of air outlets, via at least one air duct. The central arrangement of the fan in the toilet lid can be advantageous in that the installation space and the weight of the fan can be easily accommodated in the toilet lid there. Providing an air inlet there can enable air to be sucked in as directly as possible using the fan, which can then be released out of the toilet lid via the air outlet. Providing a plurality of air outlets, which can preferably be distributed as evenly as possible and/or directed towards the toilet seat or its seat surface, can improve the flow of air through the toilet interior by means of the circulating air flow.

[0044] According to a further aspect of the invention, at least the toilet mounting has a fan and an ionizer, preferably an ozone generator. In this manner, the previously described properties and functions can be implemented solely or partially by the toilet holder.

[0045] According to a further aspect of the invention, at least the toilet mounting has a neutralizer, preferably a catalyst, particularly preferably an ozone catalyst, and the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet mounting or the fan and the neutralizer of the toilet mounting. In this manner, the previously described properties and functions can be implemented solely or partially by the toilet holder.

[0046] According to a further aspect of the invention, the toilet mounting has at least one air inlet for sucking in the circulating air flow by means of the fan, and the toilet mounting has at least one air outlet for discharging the circulating air flow by means of the fan. In this manner, the previously described properties and functions can be implemented solely or partially by the toilet holder.

[0047] In this case too, the air inlet can preferably be aligned horizontally towards the toilet interior, which can bring about the advantages already described above. Preferably, the air outlet should also be aligned horizontally towards the toilet interior, which can cause the air to be released back into the middle area of the toilet interior, which can promote circulation of the air flow there. This applies accordingly to the advantages of preferably arranging the air inlet and the air outlet of the toilet mounting as far apart from one another as possible.

[0048] According to a further aspect of the invention, the toilet seat has at least one air outlet which is designed to be at least partially aligned with the air outlet of the toilet holder in the position of use of the toilet seat. This may make it possible to arrange at least the fan and preferably also the ionizer, particularly preferably also the neutralizer, in the toilet mounting, where the required installation space can be made available comparatively easily. At the same time, however, the circulating air flow can be fed to the air outlet of the toilet seat via the air outlet of the toilet mounting in order to be discharged into the toilet interior via the air outlet of the toilet seat. This can promote the circulation of airflow throughout the toilet interior.

[0049] According to a further aspect of the invention, the toilet seat has at least one duct inlet, which is connected to a duct outlet in an air-conducting manner by means of at least one flow channel, wherein the duct outlet is designed to be at least partially in line with the air inlet of the toilet holder in the position of use of the toilet seat. In this manner, the point at which air is drawn in from the toilet interior can be moved away from the mounting, which can promote the circulation of air flow through the toilet interior. Aligning the duct entrance preferably horizontally towards the toilet interior can be advantageous in that this can also promote the circulation of the air flow through the toilet interior.

[0050] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet seat and the fan and the ionizer of the toilet mounting or the fan and the ionizer of the toilet seat and the fan and the neutralizer of the toilet mounting. This allows the corresponding properties and functions described above to be implemented in these combinations. In particular, this method allows for simultaneous cleaning and/or disinfection of the toilet seat and toilet interior. Alternatively, the toilet seat can be cleaned and/or disinfected while simultaneously neutralizing the excess ionized air. This is described in more detail or in a comparable manner using the following aspect of the invention.

[0051] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet lid and the fan and the ionizer of the toilet mounting or the fan and the ionizer of the toilet lid and the fan and the neutralizer of the toilet mounting.

[0052] Accordingly, on the one hand, a first circulating air flow can be generated by means of the fan and the ionizer of the toilet lid, which can be specifically used for cleaning and/or disinfecting the toilet seat or its seat surface. At the same time, a second circulating air flow can be generated in parallel using the fan and the ionizer of the toilet mounting, which can reach the rest of the toilet interior and reduce odors there as well as clean and/or disinfect the surfaces. In this manner, the respective effects can be achieved particularly effectively, although this requires increased effort. If the toilet lid fan or its air inlet is positioned centrally in the toilet lid, this can result in a particularly effective mixing of the two circulating air streams, which can promote the most complete possible penetration of the toilet interior with ionized air.

[0053] On the other hand, the first circulating air flow can be generated by means of the fan and the ionizer of the toilet lid as described above. At the same time, in parallel, the ionized air of the first circulating air stream can be neutralized by means of the fan and its catalyst of the toilet mounting,

which can partially reach the second circulating air stream of the fan of the toilet mounting. If the fan of the toilet mounting is operated faster or more powerfully than the fan of the toilet lid, a substantially complete neutralization of the ionized air generated and emitted by the fan and ionizer of the toilet lid can occur.

[0054] According to a further aspect of the invention, the toilet bowl, the toilet seat and/or the toilet lid is/are designed to seal the toilet interior at least substantially airtight when the toilet bowl is closed by means of the toilet lid. This can be achieved by at least one suitable sealing element, which can act airtight when the toilet lid is closed in order to seal the toilet interior airtight. For this purpose, a sealing element can be used, which is arranged and acts accordingly between the toilet lid and the toilet bowl, so that the toilet seat can be enclosed radially on the outside by the sealing element. Alternatively, a suitable sealing element can be arranged between the toilet bowl rim and the toilet seat and another suitable sealing element between the toilet seat and the toilet lid, each preferably located radially on the outside, in order to also achieve this effect. In any case, it may be advantageous to arrange the toilet seat and in particular its seat surface as completely as possible within the sealing element so that the seat surface of the toilet seat can be reached by the circulating air flow for the purpose of cleaning and/or disinfection. This can also be facilitated by arranging the sealing element preferably on the rim of the toilet bowl and/or on the rim of the toilet lid. In any case, the sealing element can consist of or have a spring-elastic material in order to compensate for unevenness and improve an airtight seal. Elastomeric materials in particular can be used for this purpose.

[0055] According to a further aspect of the invention, the toilet has at least one sensor which is designed and arranged to detect the upcoming, the ongoing and/or the completed use of the toilet, wherein the toilet, preferably a control unit of the mounting, is designed and arranged to recognize the upcoming, the ongoing and/or the completed use of the toilet from the recorded sensor data of the sensor and to operate at least one fan and one ionizer in response thereto.

[0056] This allows the toilet to recognize that at least one circulating air flow should be carried out automatically as described above. This can relieve a person as a user from having to initiate the respective function themselves. In particular, this can prevent the user from forgetting to initiate the respective function.

[0057] The respective function may depend on and vary depending on whether the user is about to use the toilet, i.e. whether the user wishes to use the toilet in the near future, whether the user has already used the toilet or whether the user is about to finish using the toilet. This can be detected by the same sensor or by different sensors and recognized by a control unit by evaluating the captured sensor data.

[0058] For example, a proximity, motion or distance sensor can be used to detect and recognize that a person is moving towards the toilet, is moving within a certain distance from the toilet or is moving closer to a predetermined distance from the toilet. For this purpose, the proximity, movement or distance sensor can be arranged on the toilet bowl, on the toilet seat, on the toilet lid or on the mounting in such a manner that the proximity, movement or distance sensor can be sufficiently directed towards the surroundings of the toilet and in particular in the area in front of the toilet, in particular in the position of use of the toilet seat and/or with the toilet lid closed. Additionally or alternatively, at least one light barrier can be used for this purpose. This can be interpreted in each case as a user wishing to use the toilet, so that as a precautionary measure a circulating air flow of ionized air can be generated, preferably by a fan and ionizer of the mounting, and directed into the toilet interior, preferably below the toilet seat. This can primarily serve to reduce odors.

[0059] By means of the previously described proximity, motion or distance sensor or the previously described light barrier, the presence of a user in such close proximity to the toilet or the sensor can additionally or alternatively be detected and recognized, for example by means of correspondingly small distance values, so that it can be concluded that the user is using the toilet or sitting on the

toilet seat. Additionally or alternatively, this can also be detected by at least one force or pressure sensor or by several force or pressure sensors, preferably distributed as evenly as possible, and the use of the toilet or the user sitting on the toilet seat can be determined on the basis of the weight or the load, preferably by at least one corresponding limit value. Additionally or alternatively, a temperature sensor or several temperature sensors can be used, preferably on the seat surface of the toilet seat. This also applies to at least one brightness sensor, which can preferably be arranged inside the toilet interior. In any case, a circulating air flow of ionized air can be generated, preferably by a fan and ionizer of the mounting, and directed into the toilet interior, preferably below the toilet seat.

[0060] Likewise, by means of such sensors and similar, it can be detected and recognized through evaluation that the user has finished using the toilet or has left or left the toilet seat, for example by exceeding corresponding limit values of the distance and/or by falling below corresponding limit values of the pressure or temperature. In this case, an already flowing circulating air stream can be continued as a follow-up, preferably for a predetermined period of time, in order to reduce remaining odors and to improve the cleaning and/or disinfection of the toilet interior. Additionally or alternatively, a circulating air flow of ionized air can now be generated in the area of the toilet seat in order to clean and/or disinfect the toilet seat immediately after use. In this case, the circulating air flow can preferably be generated with sufficient strength to also transport particles from the seat surface into the opening of the toilet seat and thus remove them. This can also preferably be done for a predetermined period of time.

[0061] According to a further aspect of the invention, at least one sensor is designed and arranged to detect a presence or absence and/or an approach or departure of a user, and the toilet, preferably a control unit of the mounting, is designed and arranged to detect the presence or absence and/or the approach and/or the departure of a user from the sensor data of the sensor. This can enable a concrete implementation of previously described functions and properties of the toilet according to the invention.

[0062] According to a further aspect of the invention, at least one sensor is a force sensor or pressure sensor of the toilet bowl or toilet seat, which is designed and arranged to detect a force or a pressure depending on the presence or absence of the user. This can enable a concrete implementation of previously described functions and properties of the toilet according to the invention. The force or pressure sensor can preferably be arranged between a toilet rim of the toilet bowl and the toilet seat. In other words, in this manner, a load on the toilet seat due to the user's weight can be sensed, such that alternatively or additionally, the use of the toilet can be detected in this manner. The pressure sensor can also be considered as a force sensor.

[0063] If necessary, an elastically deformable element such as, for example, a rubber element can be arranged between the pressure sensor and the toilet bowl rim and/or the toilet seat, in order to establish contact in the vertical direction between the toilet bowl rim and the force or pressure sensor and/or between the toilet seat and the force or pressure sensor, such that forces or pressures can also be effectively transmitted and, in particular, a contact-free gap can be avoided. This can favor or ensure the functioning of the force or pressure sensor.

[0064] In any event, the detection of a use of the toilet can occur later than with a motion sensor, which is aligned to the surrounding area of the toilet or to the region above the toilet seat, since the person must have already sat down on the toilet seat to stress the pressure sensor. However, the use of the toilet can be reliably and directly detected in this manner, since motions of the user in the region around the toilet or even above the toilet interior cannot be inadvertently detected as use of the toilet without actually being one, as can be the case with a proximity, motion and/or distance sensor.

[0065] According to a further aspect of the invention, the toilet seat, the toilet lid and/or the mounting has at least one motion sensor which is designed and configured to detect the approach and/or the departure of the user. This can enable a concrete implementation of previously described

functions and properties of the toilet according to the invention. Such a motion sensor can thus detect the motion of the person and/or the user towards the toilet or towards sitting on the toilet seat on top of it and conclude from this that the toilet is about to be used. In particular, this can activate the function of reducing odors so that they can be immediately sucked away and eliminated or reduced. The motion sensor can be arranged and aligned on the toilet in such a manner that the corresponding region can be sensed. In particular, the motion sensor can be arranged and aligned in such a manner that its sensory detection range cannot be covered by a raised toilet lid. Placing the motion sensor on the outside of the toilet lid can make it possible to use the corresponding functions even when the toilet lid is closed.

[0066] According to a further aspect of the invention, the toilet seat, the toilet lid and/or the mounting has at least one distance sensor which is designed and configured to detect the presence and/or absence of the user. This can enable a concrete implementation of previously described functions of the toilet according to the invention. Placing the distance sensor on the outside of the toilet lid can make it possible to use the corresponding functions even when the toilet lid is closed.

[0067] According to a further aspect of the invention, the mounting has at least one drive unit which is designed and arranged to move the toilet lid relative to the toilet bowl in order to close or release the toilet interior. In this manner, the user can be relieved of this task. Furthermore, this may enable the toilet or a control unit of the toilet to automatically move the toilet lid to close or open, as will be described in more detail below. In any case, it may be advantageous for the drive unit to have a gear motor with a slip clutch to protect the drive unit from damage.

[0068] According to this aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to close the toilet interior by moving the toilet lid if the termination of the presence and/or the departure of the user is detected by means of the sensor data of the sensor. This allows the toilet interior to be automatically closed by means of the toilet lid, preferably airtight, so that odors are retained within the toilet interior and can be reduced or eliminated there as described above. Furthermore, this can facilitate the cleaning and/or disinfection of the toilet interior by means of circulating ionized air. Delaying the closing of the toilet interior by means of the toilet lid after the user has finished being there and/or has left can promote and/or ensure that the user has actually left the area above the toilet seat. Implementing the movement of the toilet lid preferably by pivoting and particularly preferably by vertical pivoting can enable the use of the corresponding properties and advantages described above.

[0069] Alternatively or additionally, according to this aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to release the toilet interior by moving the toilet lid if the approach of the user is detected by means of the sensor data of the sensor. This can keep the toilet closed for as long as possible by the closed toilet lid, which the user may find visually more appealing. At the same time, the toilet can then be released automatically by opening the toilet lid when a user wants to use the toilet, so that the user can be relieved of the burden of moving the toilet lid, which can also save time.

[0070] Alternatively or additionally, according to this aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to release the toilet interior by moving the toilet lid if operation of at least one fan, at least one ionizer and/or at least one neutralizer is terminated. This allows the toilet to be released automatically when the fan, ionizer and/or neutralizer have finished functioning and neither odors nor ionized air can escape into the environment when the toilet lid is opened. Preferably delaying the movement of the toilet lid from the moment of release of the toilet can ensure that these functions are completed.

[0071] Alternatively or additionally, according to this aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to release the toilet interior by not moving the toilet lid if the presence of the user is detected by means of the sensor data of the sensor. This can be considered as a safety feature not to allow the toilet lid to close the toilet as long as a user is using the toilet.

[0072] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet mounting at least for the duration of the detected presence of the user. This can ensure that the odor reduction or odor prevention function can be used at least for the duration of the user's use of the toilet. Continuing to operate this function for a predetermined period of time afterwards can ensure sufficient after-run to completely reduce the odors as safely as possible.

[0073] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the ionizer of the toilet seat, the fan and the ionizer of the toilet lid and/or the fan and the neutralizer of the toilet mounting if the termination of the presence and/or the departure of the user is detected by means of the sensor data of the sensor. This allows the corresponding functions and properties described above to be combined with one another in order to interact and/or complement one another in a particularly advantageous manner. In particular, this allows the toilet seat or its seat surface to be cleaned using ionized air and, at the same time, the ionized air can then be neutralized again using the neutralizer on the toilet mounting in order to keep the ionized air away from the area around the toilet. Doing this preferably with the toilet interior closed can increase the effectiveness of the ionized air as well as the neutralization of excess ionized air. Performing one or more of these functions preferably after a predetermined period of time and/or for a predetermined period of time can also improve the effectiveness of the function.

[0074] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and configured to operate the fan and the neutralizer of the toilet seat, the fan and the neutralizer of the toilet lid and/or the fan and the neutralizer of the toilet mounting if the termination of the presence and/or the departure of the user is detected by means of the sensor data of the sensor. This can ensure that ionized air generated during the user's presence can be kept away from the toilet environment. Doing this preferably with the toilet interior closed can increase the effectiveness of neutralizing the ionized air. Performing one or more of these functions preferably after a predetermined period of time and/or for a predetermined period of time can also improve the effectiveness of the function.

[0075] According to a further aspect of the invention, the toilet has at least one fragrance unit which is designed and configured to emit a fragrance to the environment of the toilet. In particular, at least one fragrance generator can be provided as a fragrance unit, which can be designed and positioned to release at least one fragrance and preferably alternatively at least two fragrances to the outside of the toilet. For this purpose, a storage space or a storage room of the fragrance unit, preferably within the toilet, can be provided in order to store at least one corresponding fragrance, in particular in liquid, pasty or solid form, which can be reached in particular by the air flow of a fan of the fragrance unit. This allows scent particles of the fragrance to be released into the environment and, in particular, to be carried along by the air flow of the fragrance unit's fan in order to spread the fragrance as effectively as possible in the area around the toilet. This can be combined in particular with the odor reduction function, in order to additionally provide a pleasant scent to the ambient air of the toilet, which in particular can mask any remaining unpleasant odors. Using at least two different fragrances can expand the possible uses.

[0076] According to a further aspect of the invention, the toilet, preferably a control unit of the mounting, is designed and arranged to operate the fragrance unit for dispensing the fragrance,

[0077] if the sensor data from the sensor detects the termination of the presence and/or the departure of the user and/or if the toilet interior is closed or released by moving the toilet lid.

[0078] Thus, according to the invention, a targeted fragrance release from the fragrance unit can take place in response to sensory or otherwise detected situations or conditions in the toilet.

Releasing the fragrance when the user finishes using the toilet can be beneficial in order to mask any remaining odors outside the toilet interior with the fragrance and/or to specifically scent the area around the toilet. Doing this in addition to or as an alternative to closing the toilet interior by

moving the toilet lid can be advantageous for the same reasons. Doing this additionally or alternatively depending on the opening of the toilet interior by moving the toilet lid can be advantageous in that in this case it can be assumed that the odors within the toilet interior have been reduced as completely as possible as described above and thus the fragrance can be used as completely as possible to provide the area around the toilet with the fragrance, which can be perceived as pleasant by the user instead of masking odors.

[0079] According to a further aspect of the invention, the fragrance unit comprises a fragrance cassette with a fragrance and a fragrance fan for generating an air flow, wherein the fragrance fan is designed and arranged to carry the fragrance with the air flow into the environment of the toilet. This can enable a concrete implementation of a fragrance unit as previously described. Preferably making the fragrance cassette replaceable can make it easier for the user to refill the fragrance, as the user can replace the entire fragrance cassette quickly and easily. Preferably using a liquid fragrance can facilitate the transfer of the fragrance into the air, i.e. into the gaseous form.

[0080] According to a further aspect of the invention, the toilet has at least one display element and/or a voice output, which is designed and arranged to visually display or verbally output information, preferably a cleaning and/or operating state, of the toilet to a user. This can be done in particular through symbols, colors, text and/or numbers. For this purpose, the display element can be arranged on any component of the toilet, wherein the display element can preferably be arranged on the mounting, in particular in its upper area or on top. In any case, at least one piece of information, such as a state of the toilet, a function of the toilet being performed, an energy state of an electrical energy storage device of the toilet, a fill level of a fragrance unit of the toilet and similar, can be visually displayed to the user. Alternatively or additionally, this can also be implemented by means of a voice output. This can assist the user in using the toilet or make using the toilet easier.

[0081] According to a further aspect of the invention, the toilet has at least one operating element which is designed and arranged to be actuated by a user at least to generate at least one circulating ionized air flow. The operating element can in particular be a push button, a rotary knob, a toggle switch, a button, a slider, a touch sensor or similar. Preferably, the functions of a display element and a control element can also be combined using a touch screen. In any case, a control element can enable the user to activate or deactivate at least one function of the toilet. The toilet as a whole can also be switched on and off in this way. Additionally or alternatively, at least one function of the toilet can be terminated prematurely. Additionally or alternatively, the toilet lid can also be opened or closed using the corresponding drive unit. Additionally or alternatively, the fragrance unit can also be activated or triggered in this way. This can extend the functionality of the toilet for the user.

[0082] According to a further aspect of the invention, the toilet seat, the toilet lid and/or the mounting, preferably and/or the fragrance unit, is/are designed and configured to be operated by means of an electrical energy storage device. This can preferably be done by means of a rechargeable and/or removable electrical energy storage device. Such an electrical energy storage device can be implemented as a battery that, in particular, can be inserted into the toilet device in a charged state and removed from the toilet device by the user in a discharged state, in particular by opening a flap or similar. Such an electrical energy storage device can also be designed to be rechargeable and can be permanently arranged in the toilet device in order to be recharged there, for example, by connecting a charging cable. However, a rechargeable electrical energy storage device can also be removed from the toilet device for recharging in a manner comparable to the battery described previously, and then reinserted when recharged. This applies accordingly to each of the components of the toilet. In any case, this allows electrical energy to be supplied without the need for cables to be routed to an electrical socket or similar. This can in particular reduce or avoid the wiring effort if the electrical energy storage device and the electrical consumers to be fed by it, such as a fan, an ionizer, a neutralizer, a control unit and similar, are arranged in a movable

component of the toilet such as the toilet seat or the toilet lid.

[0083] The present invention also relates to toilet seat for use in toilet as described above. In this manner, a toilet seat can be provided in order to be able to utilize its properties, functions and advantages in a toilet application according to the invention.

[0084] The present invention also relates to a toilet lid for use in a toilet as described above. In this manner, a toilet lid can be provided in order to be able to utilize its properties, functions and advantages in a toilet application according to the invention.

[0085] The present invention further relates to a mounting for use with a toilet as described above. In this manner, a mounting can be provided in order to be able to use its properties, functions and advantages in a toilet application according to the invention.

[0086] The present invention further relates to a fragrance unit for use in a toilet as described above. In this manner, a fragrance unit can be provided in order to be able to utilize its properties, functions and advantages in a toilet application according to the invention.

Description

[0087] An exemplary embodiment and further advantages of the invention are illustrated in a purely schematic manner and in detail below in connection with the following figures. Shown are:

[0088] FIG. 1 shows a perspective schematic representation of a toilet according to the invention with a closed toilet interior from an angle at the top left front;

[0089] FIG. 2 shows the representation of FIG. 1 with the toilet seat in the position of use and the toilet lid open;

[0090] FIG. 3 shows the representation from diagonally behind top left;

[0091] FIG. 4 shows the representation of FIG. 2 from the top right corner without the housing cover;

[0092] FIG. 5 shows the representation of FIG. 2 with the toilet seat in non-use position; and

[0093] FIG. 6 shows an alternative to the representation of FIG. 1;

[0094] The above figures are viewed in Cartesian coordinates. A longitudinal direction X is shown, which can also be referred to as depth X or as length X. A transverse direction Y, which can also be referred to as width Y, extends perpendicular to the longitudinal direction X. A vertical direction Z extends perpendicular to both the longitudinal direction X and the transverse direction Y, which can also be referred to as height Z and which corresponds to the direction of gravity. The longitudinal direction X and the transverse direction Y together form the horizontal X, Y, which can also be referred to as horizontal plane X, Y.

[0095] A toilet 1, 2, 3, 4, 5 according to the invention has as components or parts a toilet bowl 1, a toilet seat 2, a toilet lid 3, a mounting 4 and a fragrance unit 5. The toilet 1, 2, 3, 4, 5 can therefore also be called toilet system 1, 2, 3, 4, 5.

[0096] The toilet bowl 1 can also be referred to as toilet bowl 1 and is made in one piece from a ceramic material. The toilet bowl 1 consists of a ceramic key body 10, which substantially encloses a toilet interior 12 and ends in the vertical direction Z at the top with a toilet rim 11, which can also be referred to as toilet bowl rim 11. Towards the bottom or rear, the key body 10 merges into a toilet outlet 13, which is connected to a sewer system by a drain pipe (not shown) in order to drain flush water including excretions from the toilet 1, 2, 3, 4, 5.

[0097] The toilet seat 2 can also be called toilet seat 2 and is made of a plastic material. The toilet seat 2 consists of the seat body 20 as a multi-part injection-molded part, which in the position of use has a seat surface 20a upwards in the vertical direction Z. The seat body 20 encloses a through opening 21 substantially in a ring shape, so that the seat body 20 of the toilet seat 2 rests on the toilet rim 11 of the toilet bowl 1 in the position of use and the through opening 21 of the toilet seat 2 is in line with the toilet interior 12 of the key body 10 of the toilet bowl 1. The seat body 20 is

mounted on the mounting **4** via a pair of folding elements **22** in the form of hinges **22** so as to be pivotable by approximately 90° between the horizontal X, Y as the position of use and the vertical direction Z as the position of non-use.

[0098] On the edge side of the seat body **20**, a linear sealing element **23** is provided, which is made of an elastic material and is firmly attached to the seat body **20**. The sealing element **23** runs completely around the edge of the seat body **20**. The sealing element **23** is designed and arranged in such a manner that the light element **23** can act in an airtight manner with respect to the toilet rim **11** of the toilet bowl rim **1** and with respect to a lid rim **31** of the toilet lid **3** in the position of use of the toilet seat **2** and when the toilet lid **3** is closed. As a result, when the toilet lid **3** is closed, the interior space **12** of the toilet bowl **1** can be hermetically sealed from the environment. The toilet seat **2** is located completely within the hermetically sealed toilet interior **12**.

[0099] The toilet seat **2** has a duct inlet **24** centrally in the transverse direction Y and in the horizontal direction X, Y aligned towards the passage opening **21**. The duct inlet **24** is followed by two flow channels **25** running parallel on both sides in the transverse direction Y, which each open into a duct outlet **26** on the back of the toilet seat **2**, which are spaced apart from one another in the transverse direction Y. In the transverse direction Y, centrally between the two duct outlets **26**, an air outlet **27** is arranged, which points from the back of the toilet seat **2** underneath the seat body **20** to the toilet interior **12** of the toilet bowl rim **1**. The duct inlet **24**, the two flow channels **25**, the two duct outlets **26** and the air outlet **27** are formed by the seat body **20** of the toilet seat **2**.

[0100] The toilet lid **3** has a lid body **30**, which is also made of a plastic material. The toilet lid **3** substantially corresponds to the contour of the toilet seat **2**. The lid body **30** has the already mentioned lid edge **31**, which, when the toilet lid **3** is closed, extends in the vertical direction Z sufficiently towards the toilet bowl **1** to substantially enclose the edge of the toilet seat **2**.

[0101] On the front side, the toilet lid **3** has a first sensor **32a** in the form of a motion sensor **32a**, which is aligned in the area in front of and above the toilet **1, 2, 3, 4, 5** when the toilet lid **3** is closed. By means of the motion sensor **32a**, movements in this area can be detected sensorily. The toilet lid **3** further comprises a second sensor **32b** in the form of a distance sensor **32b**, which is arranged on the edge inside the lid body **30** such that the distance sensor **32b** is aligned in the area above the toilet seat **2** when the toilet lid **3** is opened or folded up. Accordingly, the presence or absence of a user on the toilet seat **2** can be detected by means of the distance sensor **32b**.

[0102] A fan **33**, which can also be referred to as a fan **33**, is arranged centrally in the toilet lid **3** or in its lid body **30**. The fan **33** is connected to the toilet interior **12** of the toilet bowl **1** via a central air inlet **35** in the middle of the lid body **30** when the toilet lid **3** is closed or folded down. The fan **33** is connected to an ionizer **34** in the form of an ozone generator **34** in an air-conducting manner, so that the fan **33** can suck in air from the toilet interior **12** of the toilet bowl **1** via the air inlet **35** and pass it through the ozone generator **34**. The ozone generator **34** is connected to a plurality of air outlets **37** via a plurality of air ducts **36**, which can also be referred to as air slots **37**. The air slots **37** extend in a slot-like manner close to the inside edge of the lid **31** and are arranged continuously in the circumferential direction of the toilet lid **3**. The air slots **37** are substantially of the same design and evenly spaced from one another and point towards the seat surface **20a** of the seat body **20** of the toilet seat **2** when the toilet lid **3** is folded down or closed.

[0103] The mounting **4** is box-shaped and is arranged fixedly on the rear edge, viewed in the longitudinal direction X, on the key body **10** of the toilet bowl **1**. The mounting **4** has a housing **40**, the interior of which is closed from behind in the longitudinal direction X by a removable housing cover **41**. As already mentioned, the toilet seat **2** is arranged on the mounting **4** so as to be vertically pivotable or vertically foldable by means of the pair of hinges **22**. In the vertical direction Z above the hinges **22** of the toilet seat **2**, a further pair of folding elements **42** in the form of hinges **42** is arranged, which connects the toilet lid **3** to the mounting **4** so as to be also vertically pivotable or vertically foldable by 90° between the horizontal X, Y and the vertical direction Z. Furthermore, a drive unit **43** in the form of a gear motor with slip clutch **43** is arranged within the housing **40** of

the mounting **4** and is connected to the toilet lid **3** on the output side, so that the toilet lid **3** can be folded or pivoted by the drive unit **43** of the mounting **4**.

[0104] A fan **44** in the form of a fan **44** is arranged within the housing **40** of the mounting **4**. An air inlet (not shown) of the fan **44** of the mounting **4** is located in alignment with the left duct outlet **26** of the seat body **20** of the toilet seat **2** when the toilet seat **2** is folded down into the position of use. The fan **44** of the mounting **4** is connected to a neutralizer **45** in the form of a catalyst **45** or an ozone catalyst **45**, so that an air flow can be generated from the fan **44** to the ozone catalyst **45**. The ozone catalyst **45** is followed by an ionizer **46** in the form of an ozone generator **46**. The outlet of the ozone generator **46** is connected in an air-conducting manner to an air outlet **47**, which is in line with the air outlet **27** of the seat body **20** of the toilet seat **2** when the toilet seat **2** is folded down into the position of use.

[0105] The mounting **4** further comprises a control unit **48**, which operates or controls all considered electronic parts and components of the toilet **1, 2, 3, 4, 5**. The mounting **4** further comprises a replaceable, rechargeable electrical energy storage device (not shown), which can electrically supply the control unit **48** as well as all other considered electronic parts and components of the toilet **1, 2, 3, 4, 5**.

[0106] On the left side of the mounting **4**, the fragrance unit **5** is provided, which is also arranged within the housing **40** of the mounting **4**. The fragrance unit **5** forms a separate space which is hermetically separated from the fan **44** of the mounting **4** by means of a fragrance duct wall **51**. The fragrance unit **5** in turn has an air inlet **53** of a fragrance duct **50**, through which air can be sucked into the fragrance duct **50** by a fragrance fan **52** from the environment when the toilet lid **3** is open and from the toilet interior **12** of the toilet bowl **1** between the toilet seat **2** and the toilet lid **3** when the toilet lid **3** is closed. Within the fragrance unit **5**, a replaceable fragrance cassette **55** with a liquid fragrance is held by a fragrance cassette holder **54** so that the air flow of the fragrance fan **52** can reach the liquid fragrance of the fragrance cassette **55** and carry it along as a gas. The air flow including the fragrance can be released via an air outlet **56** of the fragrance duct **50** in the transverse direction Y to the left to the surroundings of the toilet **1, 2, 3, 4, 5**.

[0107] The toilet **1, 2, 3, 4, 5** according to the invention can now be operated and used as follows by means of the control unit **48** of the mounting **4** and the other electronic components as previously described.

[0108] When not in use, the toilet **1, 2, 3, 4, 5** is in a rest state or in a standby state in which both the fan **33** and the ionizer **34** of the toilet lid **3** as well as the fan **44**, the ozone catalyst **45** and the ozone generator **46** of the mounting **4** are not operated. The fragrance unit **5** or the fragrance fan **52** is also not operated, although this can also be done at certain times or at certain intervals in the idle state in order to regularly provide the area around the toilet **1, 2, 3, 4, 5** with fragrance.

[0109] In this resting state, the toilet seat **2** is folded down or swiveled into the horizontal X, Y position. Likewise, in the resting state, the toilet lid **3** is closed by the drive unit **43** of the mounting **4** or pivoted into the horizontal X, Y. This can create a visually appealing impression for the user. In this manner, the motion sensor **32a** of the toilet lid **3** can also be used in the idle state to monitor the surroundings of the toilet **1, 2, 3, 4, 5** for movements and thus for the approach of a user.

[0110] If the control unit **48** of the mounting **4** detects a movement of a user towards toilet **1, 2, 3, 4, 5** from the sensor data of the motion sensor **32a** of the toilet lid **3**, the toilet lid **3** is pivoted or folded up in the vertical direction Z by means of the drive unit **43** of the mounting **4** in order to release the seat surface **20a** of the toilet seat **2** for the user. At the same time, the fan **44** and the ozone generator **46** of the mounting **4** can start operating, so that the fan **44** of the mounting **4** generates an air flow from the toilet interior **12** of the toilet bowl **1** via the duct inlet **24** of the toilet seat **2** into its left flow channel **25**, which passes through the left duct outlets **26** of the toilet seat **2** into the air inlet of the mounting **4**. The air flow is sucked into the fan **44** of the mounting **4** and output again in the direction of the ozone generator **46**, so that the air flow is enriched with ozone. The ionized air or ozone flows via the air outlet **47** of the mounting **4** back into the air outlet **27** of

the seat body **20** of the toilet seat **2** and from there into the toilet interior **12** of the toilet bowl **1**. [0111] Thus, the fan **44** of the mounting **4** generates a circulating air flow which leads ozone into the toilet interior **12** of the toilet bowl **1**, so that the ozone can be used to clean and/or disinfect the surfaces there during use of the toilet **1, 2, 3, 4, 5**. At the same time, ozone can counteract or neutralize odors and thus eliminate or at least reduce them directly at the point of origin. This operation of the toilet **1, 2, 3, 4, 5** can be maintained for the duration of use by the user, wherein the presence of the user can be detected by means of the distance sensor **32b** of the toilet lid **3**. The fragrance unit **5** can be operated independently of this.

[0112] If the user finishes using the toilet **1, 2, 3, 4, 5** and leaves the toilet seat **2**, this is detected by the control unit **48** of the mounting **4** from the sensor data of the distance sensor **32b** of the toilet lid **3**. After a predetermined time delay, the toilet lid **3** is now folded down into the horizontal X, Y by means of the drive unit **43** of the mounting **4** at the instigation of the control unit **48** of the mounting **4**, thus closing the toilet interior **12** of the toilet bowl **1**. The sealing element **23** of the toilet seat **2** acts as an airtight seal against the toilet rim **11** of the toilet bowl **1** and against the lid rim **31** of the toilet lid **3**. Meanwhile, the operation of the fan **44** and the ozone generator **46** of the mounting **4** can continue.

[0113] If the toilet interior **12** of the toilet bowl **1** is now hermetically sealed, the control unit **48** of the mounting **4** starts operating the fan **33** and the ozone generator **34** of the toilet lid **3**. This creates a further circulating air flow by drawing air into the fan **33** through its air inlet **35** in the middle of the underside of the toilet lid **3** towards the fan **33**. This air flow passes through the ozone generator **34** of the toilet lid **3**, so that this air flow is also enriched with ozone before the air flow is expelled via the air slots **37** onto the seat surface **20a** of the toilet seat **2**. This allows the seat surface **20a** of the toilet seat **2** to be cleaned and/or disinfected as completely as possible using ozone. In order to be able to transport particles from the seat surface **20a** of the toilet seat into the through opening **21** of the toilet seat **2**, the air flow of the fan **33** of the toilet lid **3** can be designed and operated with a correspondingly high level. In any case, this air flow can be sucked in again by the fan **33** of the toilet lid **3**, so that here too a circulating air flow with ozone is created.

[0114] Meanwhile, the operation of the fan **44** and the ozone generator **46** of the mounting **4** can continue. Both circulating air streams can mix because they meet in the upper area of the toilet interior **12** of the toilet bowl **1**. Alternatively, the operation of the fan **44** and the ozone generator **46** of the mounting **4** can be stopped immediately or after a time delay as soon as the user has finished using them. The fragrance unit **5** can be operated independently of this.

[0115] After a predetermined period of time has elapsed, after which a desired cleaning and/or disinfecting effect is usually achieved, the ozone generator **34** of the toilet lid **3** can be switched off by the control unit **48** of the mounting **4**. The fan **33** of the toilet lid **3** can continue to be operated in order to convey remaining ozone from the interior of the toilet lid **3** and from the space between the toilet lid **3** and the toilet seat **2** into the center of the toilet interior **12** of the toilet bowl **1**. At the same time, the fan **44** and the ozone catalyst **45** of the mounting **4** can be operated to suck the remaining ozone into the mounting **4** and have it neutralized there by the ozone catalyst **45**. This can prevent the user from experiencing any remaining ozone as being unpleasant when the toilet lid **3** is opened again. This can be done for a predetermined period of time. Then toilet **1, 2, 3, 4, 5** can return to the rest state described above. The fragrance unit **5** can be operated independently of this.

[0116] According to the invention, a cleaned and disinfected toilet **1, 2, 3, 4, 5** can always be made available to a user in this way, which can increase the hygiene of use for the user. In particular, a cleaned and disinfected seat surface **20a** of the toilet seat **2** can be provided. This can be done automatically.

LIST OF REFERENCE SIGNS (PART OF THE DESCRIPTION)

[0117] X longitudinal direction; depth; length [0118] Y transverse direction; width [0119] Z vertical direction; height [0120] X, Y horizontal, horizontal plane [0121] **1, 2, 3, 4, 5** toilet; toilet system [0122] **1** toilet bowl; toilet basin [0123] **10** bowl body [0124] **11** toilet rim; toilet bowl rim

[0125] **12** toilet interior [0126] **13** toilet outlet [0127] **2** toilet seat [0128] **20** seat body [0129] **20a** seat surface of the seat body **20** [0130] **21** through-opening [0131] **22** folding elements; hinges [0132] **23** sealing element [0133] **24** duct inlet [0134] **25** flow channel [0135] **26** duct outputs [0136] **27** air outlets [0137] **3** toilet lid [0138] **30** lid body [0139] **31** lid rim [0140] **32a** first sensor, motion sensor [0141] **32b** second sensor, distance sensor [0142] **33** fan; ventilator [0143] **34** ionizer, ozone generator [0144] **35** air inlet [0145] **36** air ducts [0146] **37** air outlets; air slots [0147] **4** mounting [0148] **40** housing [0149] **41** housing cover [0150] **42** folding elements; hinges [0151] **43** drive unit; gear motor with slip clutch [0152] **44** fan; ventilator [0153] **45** neutralizer, catalyst; ozone catalyst [0154] **46** ionizer, ozone generator [0155] **47** air outlet [0156] **48** control unit [0157] **5** fragrance unit [0158] **50** fragrance duct [0159] **51** fragrance duct wall [0160] **52** fragrance fans [0161] **53** air inlet of the fragrance duct **50** [0162] **54** fragrance cassette receptacle [0163] **55** fragrance cassette [0164] **56** air outlet of the fragrance duct **50**

Claims

1. A toilet comprising: a toilet bowl having a toilet interior, a mounting arranged fixedly on the toilet bowl, a toilet seat movable between a use position and a non-use position, arranged on the mounting or formed by the toilet bowl, and a toilet lid movable on the mounting to close or release the toilet interior, wherein at least one fan is arranged to generate a circulating air flow within the toilet interior of the toilet bowl, and wherein at least one ionizer is arranged to generate ionized air and to deliver it into the circulating air flow.
2. The toilet according to claim 1, wherein at least the toilet seat has a fan and an ionizer.
3. The toilet according to claim 2, wherein at least the toilet seat comprises a neutralizer, preferably a catalyst, particularly preferably an ozone catalyst, wherein the toilet comprises a control unit arranged to operate the fan and the ionizer of the toilet seat or the fan and the neutralizer of the toilet seat.
4. The toilet according to claim 2, wherein the toilet seat has at least one air inlet, preferably aligned horizontally towards the toilet interior, in order to suck in the circulating air flow by means of the fan, wherein the toilet seat has at least one air outlet, preferably directed towards the toilet lid, in order to discharge the circulating air flow by means of the fan, and wherein the air inlet and the air outlet of the toilet seat are arranged as far apart from one another as possible.
5. The toilet according to claim 1, wherein at least the toilet lid has a fan and an ionizer.
6. The toilet according to claim 5, wherein at least the toilet lid comprises a neutralizer, preferably a catalyst, particularly preferably an ozone catalyst, wherein the toilet comprises a control unit configured to operate the fan and the ionizer of the toilet lid or the fan and the neutralizer of the toilet lid.
7. The toilet according to claim 5, wherein the toilet lid has at least one air inlet, preferably aligned horizontally towards the toilet interior, in order to suck in the circulating air flow by means of the fan, and wherein the toilet lid has at least one air outlet, preferably directed towards the toilet seat, particularly preferably towards the seat surface, in order to discharge the circulating air flow by means of the fan, wherein the air inlet and the air outlet of the toilet lid are preferably arranged as far apart from one another as possible.
8. The toilet according to claim 7, wherein the toilet lid has a plurality of air outlets which are directed towards the toilet seat, particularly preferably towards the seat surface, and are preferably arranged in a uniformly distributed manner.
9. The toilet according to claim 7, wherein the fan is arranged centrally in the toilet lid, is directly connected to the air inlet and is connected in an air-conducting manner to at least the air outlet, preferably to the plurality of air outlets, via at least one air duct.
10. The toilet according to claim 1, wherein at least the toilet mounting has a fan and an ionizer.
11. The toilet according to claim 10, wherein at least the toilet mounting has a neutralizer,

preferably a catalyst, particularly preferably an ozone catalyst, wherein the toilet comprises a control unit configured to operate the fan and the ionizer of the toilet mounting or the fan and the neutralizer of the toilet mounting.

12. The toilet according to claim 10, wherein the toilet mounting has at least one air inlet, preferably aligned horizontally towards the toilet interior, in order to suck in the circulating air flow by means of the fan, and wherein the toilet mounting has at least one air outlet, preferably aligned horizontally towards the toilet interior, in order to discharge the circulating air flow by means of the fan, wherein the air inlet and the air outlet of the toilet mounting are preferably arranged as far apart from one another as possible.

13. The toilet according to claim 12, wherein the toilet seat has at least one air outlet at least partially in line with the air outlet of the toilet holder in the position of use of the toilet seat.

14. The toilet according to claim 12, wherein the toilet seat has at least one duct inlet, preferably aligned horizontally towards the toilet interior, which is connected to a duct outlet by means of at least one flow channel in an air-conducting manner, wherein the duct outlet is designed to be at least partially aligned with the air inlet of the toilet holder in the position of use of the toilet seat.

15. The toilet according to claim 2, wherein the toilet comprises a control unit configured to operate the fan and the ionizer of the toilet seat and the fan and the ionizer of the toilet mounting or the fan and the ionizer of the toilet seat and the fan and the neutralizer of the toilet mounting.

16. The toilet according to claim 5, wherein the toilet comprises a control unit configured to operate the fan and the ionizer of the toilet lid and the fan and the ionizer of the toilet mounting or the fan and the ionizer of the toilet lid and the fan and the neutralizer of the toilet mounting.

17-36. (canceled)

37. The toilet of claim 1, wherein the at least one ionizer comprises an ozone generator arranged to generate ozone and deliver it into the circulating air flow.
